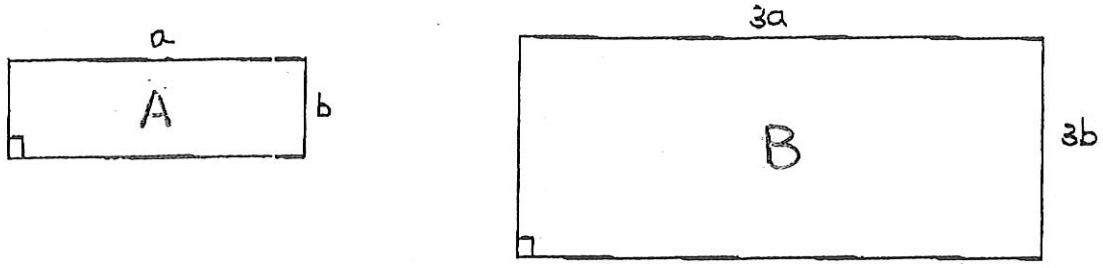
Scale factor investigation

Similar shapes

The following shapes are similar

1)

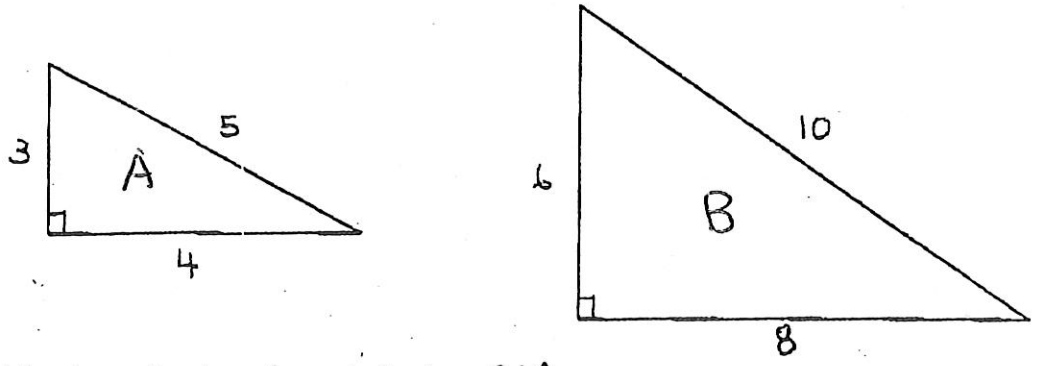


a) Write down the length scale factor B : A =

b) Write down the area scale factor:

$$\frac{\text{Area } B}{\text{Area } A} = _____$$

2)

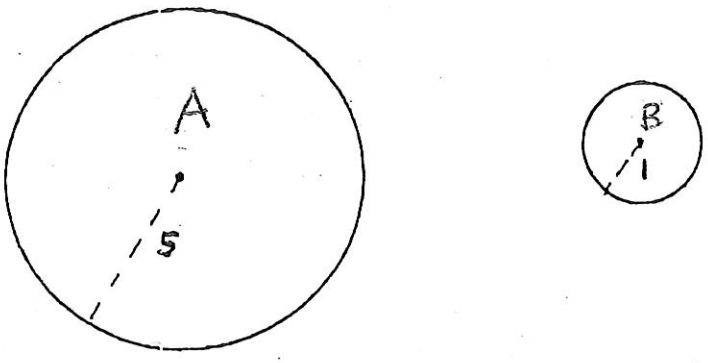


a) Write down the length scale factor B : A = _____

b) Write down the area scale factor:

$$\frac{\text{Area } B}{\text{Area } A} = _____$$

3)

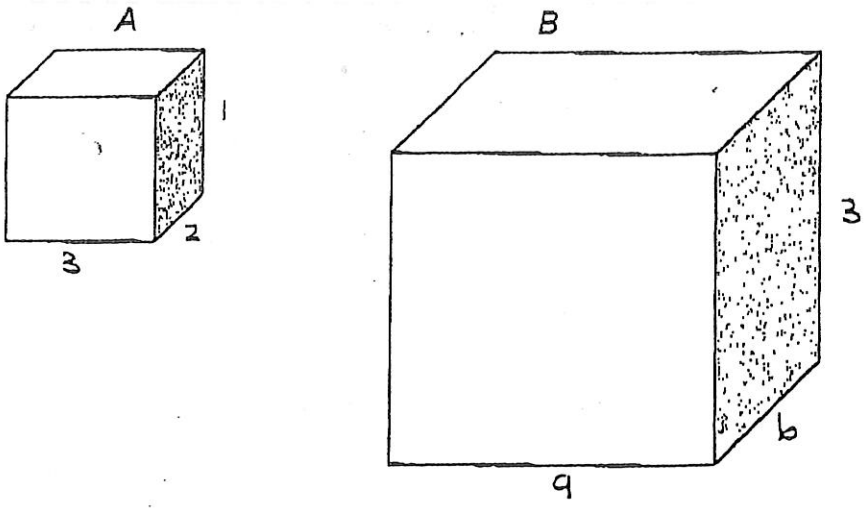


a) Write down the length scale factor A : B : = _____

b) Write down the area scale factor:

$$\frac{\text{Area } A}{\text{Area } B} = _____$$

4)



a) Write down the length scale factor B : A = _____

b) Find the surface area of the small box | Find the surface area of the large box

c) Find the area scale factor:

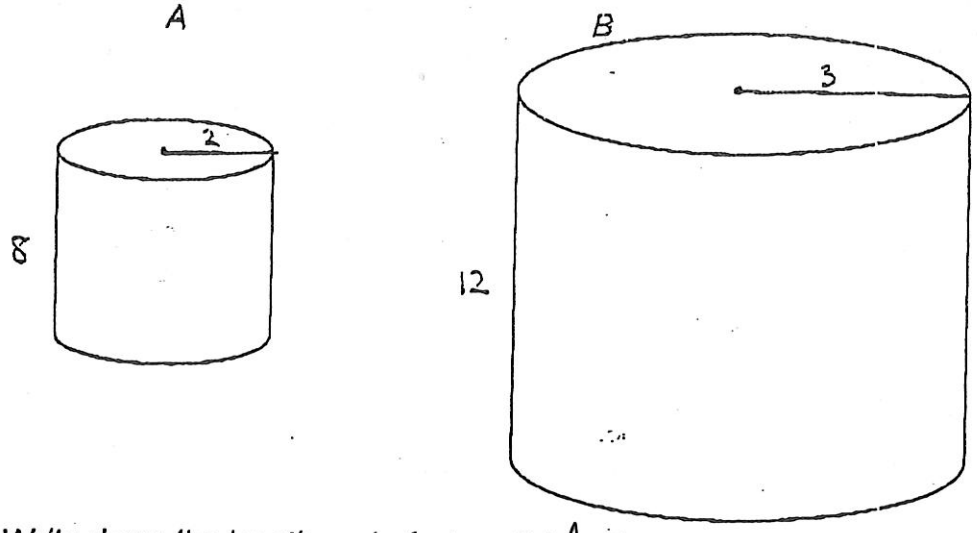
$$\frac{\text{Area of large box (B)}}{\text{Area of small box (A)}} = _____$$

d) Find the volume of the small box(A) | Find the volume of the large box (B)

e) Therefore, the volume scale factor :

$$\frac{\text{Volume of large box (B)}}{\text{Volume of small box (A)}} = _____$$

5)



a) Write down the length scale factor B : A = _____

b) Find the surface area of the small cylinder — Find the surface area of the large cylinder

c) Find the area scale factor:

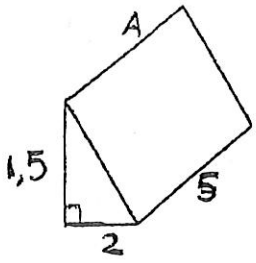
$$\frac{\text{Area of large cylinder (B)}}{\text{Area of small cylinder (A)}} = _____$$

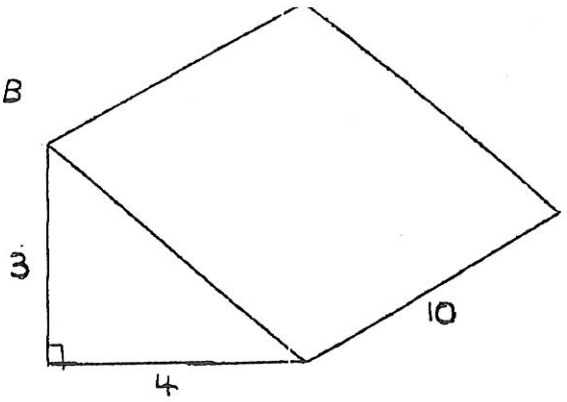
) Find the volume of the small cylinder(A) — Find the volume of the large cylinder(B)

e) Therefore, the volume scale factor :

$$\frac{\text{Volume of large cylinder (B)}}{\text{Volume of small cylinder (A)}} = _____$$

6)





a) *Write down the length scale factor* B : A

b) *Find the surface area of each triangular prism*

c) $\frac{\text{Area of large triangular prism (B)}}{\text{Area of small triangular prism (A)}}$ = _____

d) *Find the volume of both triangular prisms*

e) *Therefore, the volume scale factor =*

$\frac{\text{Volume of large triangular prism (B)}}{\text{Volume of small triangular prism (A)}}$ = _____

Now make a table of all the results:

Shape	1	2	3	4	5	6
Length scale factor						
Area scale factor						
Volume scale factor						

In general:

Length scale factor k

Area scale factor _____

Volume scale factor